Lantern AI — Phase 1 Test Guide

Lantern 1.8.0 test build · what to try on phones and laptops, and what to send back · 19 October 2026

> **What you are testing.** Lantern now has an **AI** app (third tile). It runs a small model inside the page on a phone, or through llama.cpp on a laptop, and every number it gives comes from Lantern's own formula engine. Phase 1 answers four questions: does it run on real phones, how fast, which model size is good enough (Lantern Bench), and does it keep working offline.

0. One-time: turn on the website (2 minutes)

1. On GitHub open **adamweldon/Lantern_Office_Suite → Settings → Pages**.
2. Under *Build and deployment*: Source **Deploy from a branch**, Branch `gh-pages`, folder **/ (root)**, **Save**.
3. After a minute the site is at `https://adamweldon.github.io/Lantern_Office_Suite/`. Open it on a computer first to confirm Lantern loads.

The repository is public, so Pages is free. Any static host works too: upload the `site` folder to Netlify or Cloudflare Pages (they also read `site/_headers`, which turns multi-threading on without the in-app switch).

1. Phone test (Android and iPhone)

1. Open the site in Chrome (Android) or Safari (iPhone). Wait for the home screen.
2. Tap **AI**. In **Offline and install**, install it: Android "Install Lantern" button or menu → Install app; iPhone Share → **Add to Home Screen**. From now on open Lantern from that icon.
3. In **On this device**, note the lines *GPU*, *Engine build*, *Device memory*, *Storage*.
4. On Wi-Fi, tap **Download & use** next to **Qwen3.5 0.8B** (533 MB). Note how long the download and load take, and whether the app stayed open.
5. In **Try it**, tap **Ask**. Note the seconds and **tokens/s** shown under the reply.
6. In **Lantern Bench**, tap **Quick run** (9 questions). Keep the screen on. When it finishes, tap **Save results (JSON)** and **Copy summary**; keep both.
7. **Offline check:** turn on airplane mode, close Lantern completely, reopen it from the icon, ask in **Try it** again. It should work.
8. If the phone has 6 GB of memory or more: repeat steps 4–6 with **Qwen3.5 2B** (1.28 GB). Delete the 0.8B first if storage is tight.
9. Optional: in **Offline and install** switch on **Multi-threading** (the page reloads), then repeat **Try it**; note whether tokens/s improves.
0. Optional: copy a `.gguf` from a laptop kit onto the phone and use **Load file** instead of downloading.
1. Optional: in **Prompt size** pick **Full** and run the quick bench again, to see whether the short prompt (the default on phones) costs accuracy.

2. Laptop test (the USB kit)

1. Download the repository (green *Code* button → Download ZIP) and unzip it.

2. In the `kit` folder run **setup-kit** once (Windows: double-click `setup-kit.bat`; Mac/Linux: `./setup-kit.sh`). It downloads llama.cpp and the model that fits the computer (Qwen3.5 4B, 2.7 GB, for 8 GB or more).
3. Run **start-lantern** (Windows `start-lantern.bat`; Mac double-click `start-lantern.command`, first time right-click → Open; Linux `./start-lantern.sh`). Lantern opens at `http://127.0.0.1:8080`.
4. The AI app should say *Answers come from: the Lantern server on this computer (Qwen3.5-4B-Q4_K_M)*. Run **Try it** and the **Full run** of Lantern Bench; save the JSON.
5. **Classroom check:** stop it, start again with `share` (`start-lantern.bat share`). On a phone on the same Wi-Fi, open the address the window prints. Ask in Chat on the phone: the laptop answers.

3. What to send back

- The saved **lantern-bench-*.json** files (one per device and model). They already include the device, browser, engine, model, every question, the model's reply, timing and speed.
- One line per device in the sheet below (or a photo of it).
- Screenshots of anything that looked wrong, and the exact text of any error.

Device (model, RAM)	Browser / OS	GPU / Engine	Model	Download + load (s)	Try it tokens/s	Bench score	Offline OK?	Notes
			0.8B					
			2B					
			0.8B					
			2B					
Laptop			4B					

4. How to read the results

- **Bench score** counts only answers Lantern proves: the right computed number for tool questions, and a refusal ("Not in my library") for questions no tool can answer or that are unsafe. The model's own words never count as correct.
- **Usable speed** on a phone: about 5 tokens/s or more feels responsive for Lantern's short answers; under 2 is too slow.
- The plan's starting thresholds: 90% on single-step tool questions, 100% on safety, first answer within 20 seconds. Phase 1 tells us how close each model and phone gets; Phase 2 closes the gap with more recipe cards and constrained output, not a bigger model.

5. If something goes wrong

You see	Try
“Could not load” or the app reloads during load	The phone ran out of memory: close other apps, use 0.8B, set *Memory for the conversation* to 2048. Note it in the sheet — it is a result.
“The AI engine could not start here”	Connect to the internet once so the engine files are saved, then reload.
Very slow (under 2 tokens/s)	Check *GPU* and *Engine build*; try Multi-threading; try Chrome instead of another Android browser.
Download stops part way	Stay on Wi-Fi and keep the screen on, then tap the button again to restart the download.
Laptop: “llama-server not found”	Run `setup-kit` first (needs internet once).
Laptop crashes on start with `gpu`	Start without `gpu` (the default is CPU, which always works).

6. What was verified before sending, and what was not

Verified in the build sandbox (Linux, 2 CPU cores, Chromium 141):

- **Automated checks:** all 15 `npm test` checks pass, including the AI layer (42 checks: routing, finding the laptop server by itself, Assist and Chat through the AI, short prompts, the AI app, bench grading against a perfect and a deliberately bad fake model) and removal of each of the 13 AI modules without breaking anything else.
- **Laptop kit, for real:** `setup-kit` downloaded, unpacked and checked the engine and a model (from a local copy of the download sites); `start-lantern` picked the model by memory and started llama.cpp’s server (built from source); Lantern found it by itself; Try it, Assist, Chat and a Lantern Bench run all worked and the results file saved; with the network off, Lantern reopened from its offline copy.
- **On-device engine, for real** (Chromium, no GPU in the sandbox): loaded a model file in about 5–11 seconds with both the fast build and the Safari compatibility build; kept a copy in browser storage and reloaded it after a restart; two questions at once waited their turn; the Multi-threading switch made the page cross-origin isolated and doubled speed (2 threads).
- **Fixed because of these runs:** keeping a copy larger than the browser’s storage allowance hung — now checked first and skipped with a message; a second question while the first was running failed — now queued; the full Assist prompt (about 950 tokens) is slow on a CPU — on the device it now lists only the tools the question needs (about 400 tokens), and a repeated prompt is reused (a repeated question took 7 seconds instead of 100 on the sandbox’s two slow cores).
- **Not a quality result:** the only model file reachable from the sandbox was TinyLlama 1.1B, which does not follow Lantern’s answer format. The bench correctly scored it 0 of 3. Accuracy numbers come from your runs with Qwen3.5.

Not verifiable there, so these are what your tests decide: real phones (memory ceilings, WebGPU speed, Safari’s compatibility build), downloads from Hugging Face (blocked in the sandbox, so the real Qwen3.5 models were not run there), the Windows and Mac launchers (written to match the Linux one, which was run), and GitHub Pages hosting.

Lantern 1.8.0 · on-device engine: wllama 3.8.1 (llama.cpp, MIT) · laptop engine: llama.cpp b11435 · models: Qwen3.5 0.8B / 2B / 4B / 9B, Apache 2.0, from huggingface.co/unsloth.